



THE UNIVERSITY
of EDINBURGH



Biotechnology and
Biological Sciences
Research Council



THE ROYAL
SOCIETY

Theory: Autopolyploids

Chris Gaynor

Minneapolis, June 2026

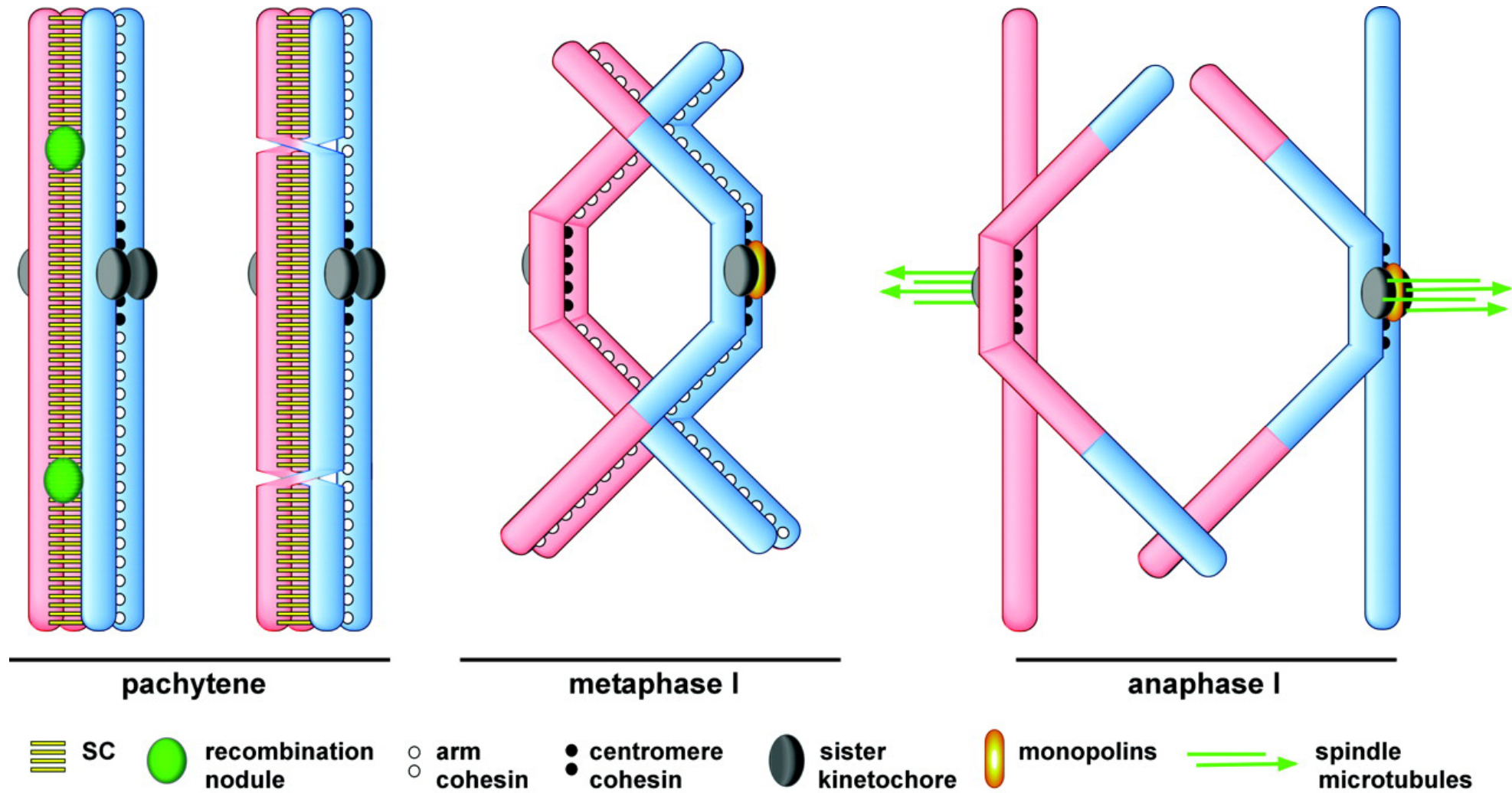


Bayer CropScience

Types of Polyploids

- Paleopolyploids (e.g. maize)
 - Genome is a fusion of ancestral genomes
 - Diploids with large chromosomes and gene duplications
- Allopolyploids (e.g. wheat)
 - Genome contains subgenomes from related species
 - Diploid like inheritance as subgenomes don't recombine
- Autopolyploids (e.g. potato)
 - More than two copies of each chromosome
 - Complex meiosis

Meiosis Review



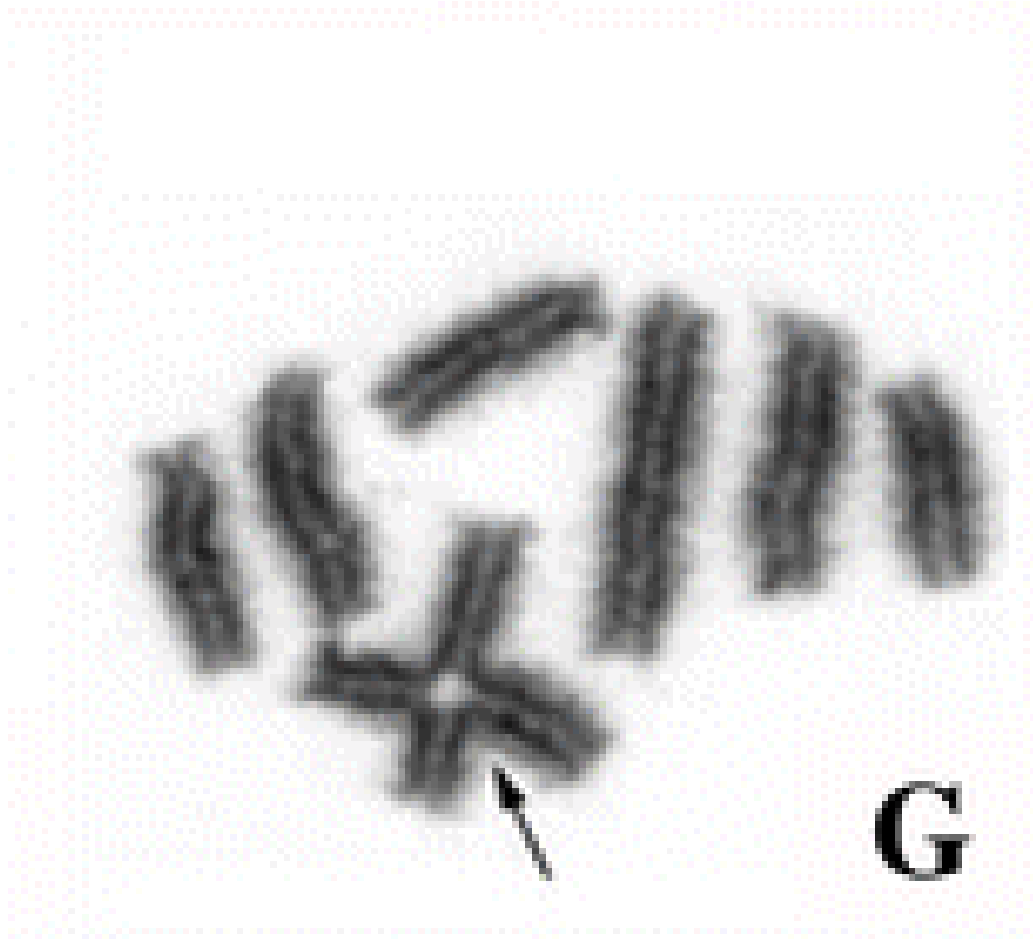
Source: Page and Hawley (2003)

Tetraploid Chromosome Paring



Figure 1 Three metaphase configurations of one set of homologous chromosomes in tetraploid meiosis. **A.** two bivalents; **B.** "cross-type" quadrivalent; **C.** "parallel" quadrivalent. In a cross-type quadrivalent two branches contain the tops of two chromosomes, and two branches the bottoms of two chromosomes; the location of the chromosome exchange point (the position where the branches meet) may vary between meioses.

Example Images



Important Considerations

- Chromosome pairing is species specific
 - Ratio of bivalent to quadrivalent pairing
 - AlphaSimR model cross-type quadrivalents or bivalents
- Location of centromere matters in quadrivalent pairing
 - AlphaSimR defaults to middle of chromosome (metacentric)
 - Determines probability of a double reduction
- Random mating does not achieve HWE

Double Reduction

- It's possible to get two copies from a single chromosome
 - Only occurs in quadrivalent pairing
 - Resulting in higher rates of inbreeding
- Assume a tetraploid with four alleles (ABCD)
 - Potential gametes with bivalent pairing:
 - AB, AC, AD, BC, BD, CD
 - Additional gametes possible with quadrivalent pairing:
 - AA, BB, CC, DD

Genotype Coding

- Universal in diploids, but no consensus in polyploids
- Higher orders of dominance (polynomial series)
 - Digenic, trigenic, tetragenic

Diploid		Tetraploid (Easton, 1976)	
Dosage	Value	Dosage	Value
2	a	4	2a
		3	a+3d+t+w
1	d	2	4d
		1	-a+3d-t+w
0	-a	0	-2a

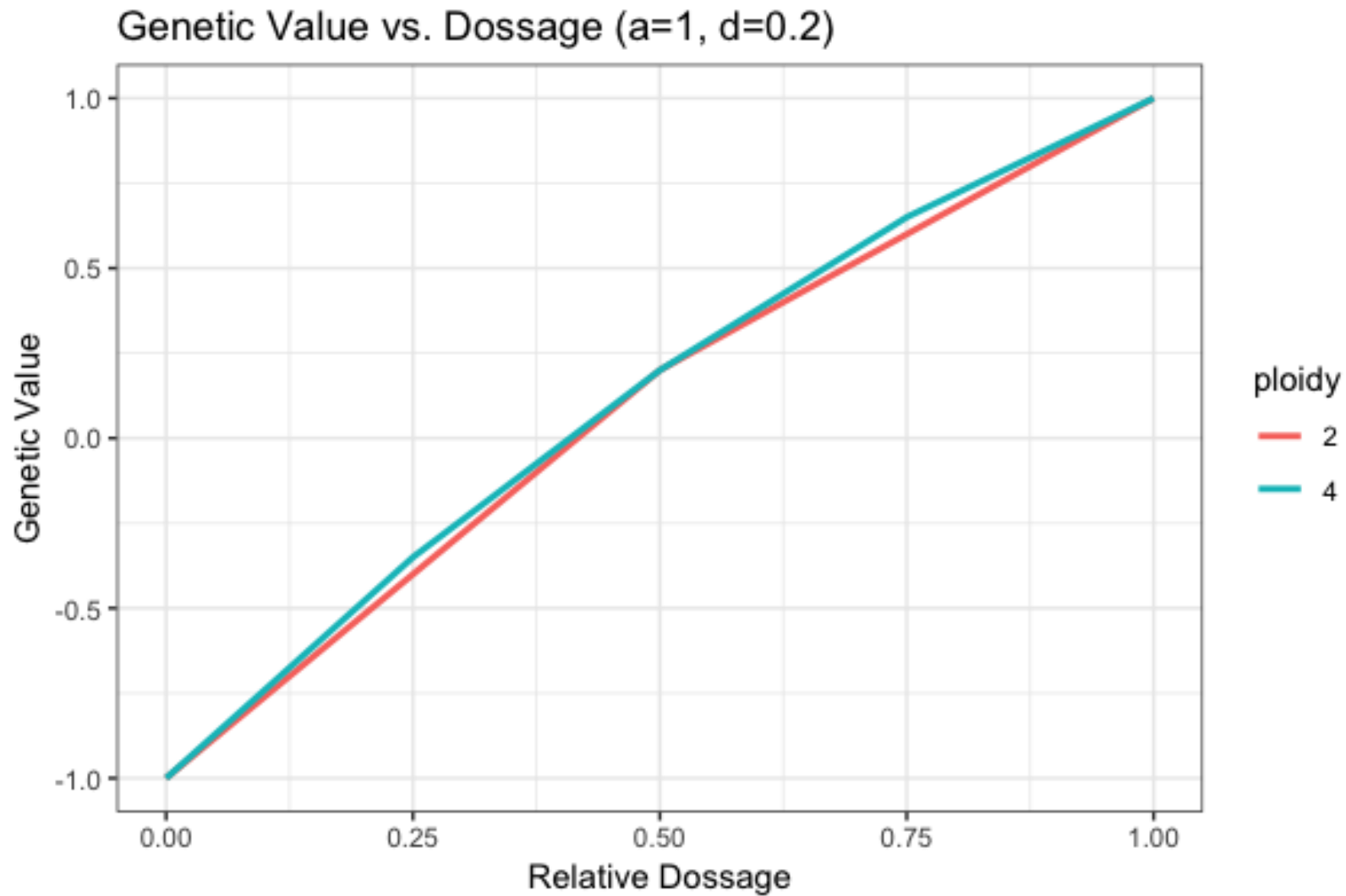
AlphaSimR Coding (Digenic Only, Dosage Scaling)

Relative dosage	Diploid	Tetraploid	Value
1	2	4	a
0.75		3	0.5a+0.75d
0.5	1	2	d
0.25		1	-0.5a+0.75d
0	0	0	-a

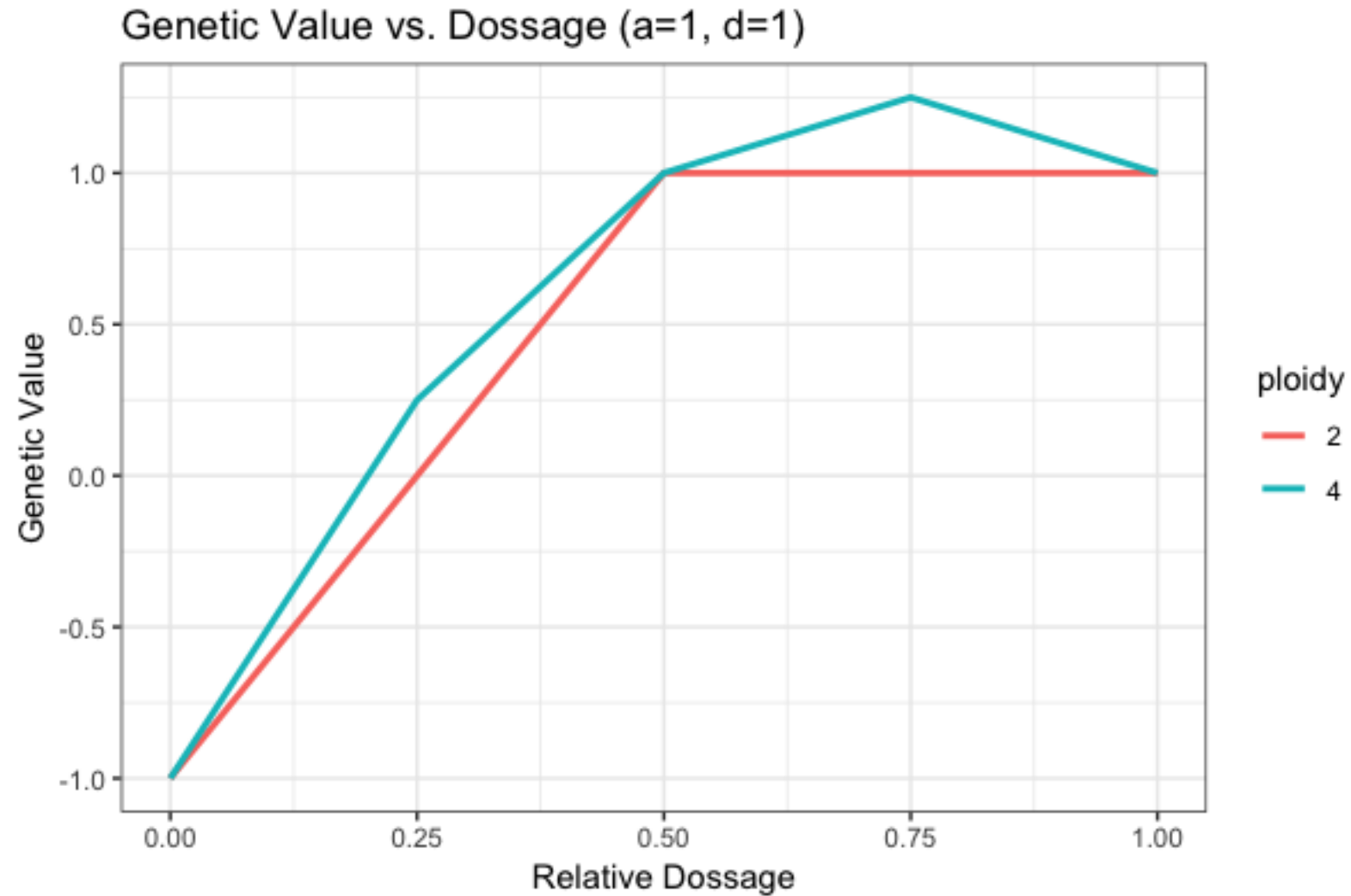
$$a = \left(x - \frac{ploidy}{2}\right) \left(\frac{2}{ploidy}\right)$$

$$d = x(ploidy - x) \left(\frac{2}{ploidy}\right)^2$$

Partial Dominance



“Complete” Dominance



Unique Polyploid Properties

- Griffing like effect on dominance variation
 - Digenic dominance contributes to selection response in tetraploids
 - Progeny inherit two chromosome copies from their parents
 - Should extend to higher order dominance at higher ploidy
- Progressive heterosis
 - Heterosis is maximized with a single cross in diploids ($A \times B$)
 - Double cross needed in tetraploids $(A \times B) \times (C \times D)$

AlphaSimR Demonstration